Set 28: Summarizing Relationships Between Categorical Features

Skill 28.1: Create and interpret contingency tables
Skill 28.2: Create and interpret an expected contingency table
Skill 28.3: Calculate and interpret the Chi-Square statistic to assess the association between two categorical variables

Skill 28.1: Create and interpret contingency tables

Skill 28.1 Concepts

In this lesson, we will cover ways of examining an association between two categorical variables

As an example, we'll explore a sample of data from the Narcissistic Personality Inventory (NPI-40), a personality test with 40 questions about personal preferences and self-view. There are two possible responses to each question. The sample we'll be working with contains responses to the following,

- influence: yes = I have a natural talent for influencing people; no = I am not good at influencing people.
- blend_in: yes = I prefer to blend in with the crowd; no = I like to be the center of attention.
- special: yes = I think I am a special person; no = I am no better or worse than most people.
- leader: yes = I see myself as a good leader; no = I am not sure if I would make a good leader.
- authority: yes = I like to have authority over other people; no = I don't mind following orders.

npi

	influence	blend_in	special	leader	authority
0	no	yes	yes	yes	yes
1	no	yes	no	no	no
2	yes	no	yes	yes	yes
3	yes	no	no	yes	yes
4	yes	yes	no	yes	no

As you might guess, responses to some of these questions are associated. For example, if we know whether someone views themself as a good leader, we may also find that they're more likely to like having authority. In this lesson we'll learn how to assess whether an association exists between any two of these variables.

Contingency tables, also known as two-way tables or cross-tabulations, are useful for summarizing two variables at the same time. For example, suppose we are interested in understanding whether there is an association between influence (whether a person thinks they have a talent for influencing people) and leader (whether they see themself as a leader). We can use the crosstab function from pandas to create a contingency table,

Code

```
influence_leader_freq = pd.crosstab(npi.influence, npi.leader)
print(influence_leader_freq)
```

Output

```
leader        no    yes
influence
no          3015   1293
yes         2360   4429
```

This table tells us the number of people who gave each possible combination of responses to these two questions. For example, 2360 people said that they do not see themselves as a leader but have a talent for influencing people.

To assess whether there is an association between these two variables, we need to ask whether information about one variable gives us information about the other. In this example, we see that among people who **don't** see themselves as a leader (the first column), a larger number (3015) **don't** think they have a talent for influencing people. Meanwhile, among people who **do** see themselves as a leader (the second column), a larger number (4429) **do** think they have a talent for influencing people.

So, if we know how someone responded to the leadership question, we have some information about how they are likely to respond to the influence question. This suggests that the variables are associated.

Just as we saw with bar charts, it is often helpful to convert frequencies to proportions. This can be done by dividing the counts in the contingency table by an appropriate total. Dividing by the overall total gives overall proportions, while dividing by row or column totals allows us to compare conditional proportions and better assess whether an association exists.

Code

```
influence_leader_freq = pd.crosstab(npi.influence, npi.leader)
influence_leader_prop = pd.crosstab(npi.influence, npi.leader)/len(npi)
print(influence_leader_prop)
```

Output

```
leader            no       yes
influence
no          0.271695  0.116518
yes         0.212670  0.399117
```

Skill 28.1 Exercise 1

Skill 28.2: Create and interpret an expected contingency table

Skill 28.2 Concepts

In the previous exercise, we calculated the proportions for the leader and influence questions.

leader	no	yes
influence		
no	0.271695	0.116518
yes	0.212670	0.399117

To determine whether these variables are associated, we can use the marginal proportions to construct a table of expected proportions under the assumption that there is no association between them. Below explains how to do this,

Step 1: Find the Marginal Proportions

First, compute row totals and column totals.

Row Totals (Influence)

Influence = no:

0.271695 + 0.116518 = 0.388213

Influence = yes:

0.212670 + 0.399117 = 0.611787

Column Totals (Leader)

Leader = no:

0.271695+0.212670=0.484365

Leader = yes:

0.116518+0.399117=0.515635

leader	no	yes	row totals
influence			
no	0.271695	0.116518	0.388213
yes	0.212670	0.399117	0.611787
column totals	0.484365	0.515635	

Step 2: Calculate Expected Proportions (Assuming Independence)

If the variables are independent:

Expected proportion=(row proportion)×(column proportion)

	Actual Proportion				Calculated Expected Proportion (no association)		
leader	no	yes	row totals		leader	no	yes
influence					influence		
no	0.271695	0.116518	0.388213		no	0.484*0.388 = 0.188	0.516*0.388 = .200
yes	0.212670	0.399117	0.611787		yes	0.484*0.612 = 0.296	0.516*0.612 = 0.315
column totals	0.484365	0.515635					

These proportions can then be converted to frequencies by multiplying each one by the sample size (11097 for this data):

	Actual Counts			Calculated Expected Counts (no association)		
leader	no	yes		leader	no	yes
influence				influence		
no	3015	1293		no	0.188*11097 = 2087	0.200*11097 = 2221
yes	2360	4429		yes	0.296*11097 = 3288	0.315*11097 = 3501

The table above tells us that if there were no association between the leader and influence questions, we would expect 2087 people to answer no to both.

In python, we can calculate this table using the *chi2_contingency()* function from *SciPy*, by passing in the observed frequency table. There are actually four outputs from this function, but for now, we'll only look at the fourth one, Below illustrates how we can combine all these steps with one function,

npi

	influence	blend_in	special	leader	authority
0	no	yes	yes	yes	yes
1	no	yes	no	no	no
2	yes	no	yes	yes	yes
3	yes	no	no	yes	yes
4	yes	yes	no	yes	no

Code

```
# Creates a cross table of actual counts
contingency_table = pd.crosstab(npi['influence'], npi['leader'])

# Creates a table of expected counts
chi2, p, dof, expected = chi2_contingency(contingency_table)

# Displays the expected counts
display(np.round(expected))
```

Output

```
array([[2087., 2221.],
       [3288., 3501.]])
```

Now that we have the expected contingency table (no association), we can compare it to our observed contingency table,

	Actual Counts			Calculated Expected Counts (no association)		
leader	**no**	**yes**		leader	no	yes
influence				influence		
no	3015	1293		no	2087	2221
yes	2360	4429		yes	3288	3501

The more that the expected and observed tables differ, the more sure we can be that the variables are associated. In this example, we see some pretty big differences (eg., 3015 in the observed table compared to 2087 in the expected table). This provides additional evidence that these variables are associated.

[**Skill 28.2 Exercises 1 & 2**]

Skill 28.3: Calculate and interpret the Chi-Square statistic to assess the association between two categorical variables

Skill 28.3 Concepts

In the previous exercise, we calculated a contingency table of expected frequencies if there were no association between the leader and influence questions. We then compared this to the observed contingency table. Because the tables looked somewhat different, we concluded that responses to these questions are probably associated.

While we can inspect these tables visually, many data scientists use the *Chi-Square statistic* to summarize **how** different these two tables are. To calculate the Chi Square statistic, we simply find the squared difference between each value in the observed table and its corresponding value in the expected table, and then divide that number by the value from the expected table; finally add up those numbers,

$$ChiSquare = \sum \frac{(observed - expected)^2}{expected}$$

The Chi-Square statistic is also the first output of the *SciPy* function *chi2_contingency(),*

npi

	influence	blend_in	special	leader	authority
0	no	yes	yes	yes	yes
1	no	yes	no	no	no
2	yes	no	yes	yes	yes
3	yes	no	no	yes	yes
4	yes	yes	no	yes	no

Code

```
# Creates a cross table of actual counts
contingency_table = pd.crosstab(npi['influence'], npi['leader'])

# Creates a table of expected counts
chi2, p, dof, expected = chi2_contingency(contingency_table)

# Displays the expected counts
display(np.round(expected))
# Displays the chi-squared statistic
display(np.round(chi2))
```

Output

```
array([[2087., 2221.],
       [3288., 3501.]])
np.float64(1308.0)
```

The interpretation of the Chi-Square statistic is dependent on the size of the contingency table. For a 2x2 table (like the one we've been investigating), a Chi-Square statistic larger than around 4 would strongly suggest an association between the variables. In this example, our Chi-Square statistic is much larger than that — 1307.88! This adds to our evidence that the variables are highly associated.

Skill 28.3 Exercise 1